

Course Project: Research assignment in pattern recognition

EE798R

Project Theme

Each student will select a paper related to **Pattern Recognition related areas** and prepare a detailed technical report.

The objective of the project is not only to summarize the paper, but to demonstrate a clear understanding of its **theoretical foundations, mathematical formulation, algorithmic implementation, and quantitative and qualitative results**.

The final report should read like a compact scientific report combined with a small experimental study.

1. Choice of Topic

The chosen topic may be theoretical, algorithmic, application-oriented, or based on a recent research direction.

Examples include, but are not limited to:

- Advanced supervised or unsupervised learning methods
- Self-supervised and contrastive learning
- State-space models and Mamba
- Retrieval-Augmented Generation (RAG)
- Graph Neural Networks
- Foundation models
- Vision Transformers
- Multimodal learning
- Diffusion models
- Generative models
- Time-series forecasting
- Metric learning
- Few-shot and zero-shot learning
- Open-set and out-of-distribution recognition
- Explainable AI
- Continual learning
- Domain adaptation
- Active learning
- Anomaly detection
- Imbalanced learning

- Bayesian learning
- Probabilistic graphical models
- AI/ML applications in vision, speech, healthcare, remote sensing, robotics, etc.

2. Mandatory Structure of the Report

The report should contain the following sections.

2.1. Title and abstract

The title should clearly indicate the selected topic. The abstract should briefly summarize the problem, method or family of methods studied, experimental setup, and major observations or conclusions.

2.2. Introduction and motivation

Explain the problem being addressed, why it is important, where the method is useful, major challenges, and the scope of the report.

2.3. Background and related work

Provide the necessary background for understanding the selected topic. Students should discuss important prior methods and clearly explain how the chosen method/topic relates to existing approaches. A simple chronological listing of papers is not sufficient.

2.4. Theory and conceptual explanation

This section should provide a detailed explanation of the core ideas behind the selected topic. The explanation should be written in the student's own words and should include intuition, assumptions, model architecture or algorithmic pipeline, important design choices, advantages, and limitations. Relevant diagrams or flowcharts are strongly encouraged.

2.5. Mathematical formulation

This is a mandatory section. Students should explain the mathematical formulation of the selected paper. Depending on the topic, this may include objective functions, probability models, loss functions, optimization objectives, matrix/vector formulations, derivations of important equations, update rules, complexity analysis, and probabilistic or geometric interpretation.

Equations should not simply be copied from papers. Important symbols and terms must be defined and explained.

For example,

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(x_i), y_i), \quad (1)$$

should be followed by an explanation of the role of f_{θ} , the loss $\ell(\cdot)$, the parameters θ , and the samples (x_i, y_i) .

2.6. Algorithm / Methodology

Provide a clear description of the complete algorithm or model pipeline. This section should include, wherever applicable, input and output, preprocessing, feature representation, model architecture, training procedure, inference procedure, optimization method, and important hyperparameters. Pseudocode may be included if useful.

2.7. Implementation and code snippets

The report must contain meaningful code snippets illustrating important parts of the implementation, such as model definition, preprocessing, loss implementation, training loop, inference pipeline, or evaluation code.

```
optimizer.zero_grad()
prediction = model(x)
loss = criterion(prediction, y)
loss.backward()
optimizer.step()
```

Listing 1: Illustrative training step

Students should explain what the code is doing. Large blocks of copied code should be avoided.

2.8. Dataset and experimental setup

Clearly describe the dataset(s) used, number of samples, train/validation/test split, preprocessing, evaluation protocol, hardware/software setup, hyperparameters, and baseline methods used for comparison.

2.9. Quantitative results

The report must include quantitative evaluation. Suitable metrics may include accuracy, precision, recall, F1-score, ROC-AUC, MSE, MAE, PSNR, SSIM, mAP, IoU, retrieval metrics, runtime, memory usage, parameter count, or FLOPs, depending on the problem.

At least one clearly formatted results table should be included.

Table 1: Example quantitative comparison.

Method	Metric 1	Metric 2	Runtime
Baseline 1	—	—	—
Baseline 2	—	—	—
Proposed / Reproduced Method	—	—	—

Students should interpret the values instead of presenting the table without discussion.

2.10. Qualitative results

Qualitative analysis is mandatory whenever applicable. Examples include prediction visualizations, generated images, segmentation outputs, attention maps, confusion examples, retrieval examples, feature-space visualizations, reconstructed signals, and failure cases. Each figure should have a meaningful caption and be discussed in the text.

2.11. Analysis and discussion

Students should discuss why the method performs well or poorly, comparison with baselines, impact of important hyperparameters, observed failure cases, computational requirements, limitations, and situations where the method may not work. Whenever possible, an ablation or sensitivity analysis is encouraged.

2.12. Conclusion

Summarize what was learned, major experimental observations, strengths and limitations, and possible future directions.

2.13. References

All papers, books, websites, datasets, repositories, and external resources used in the report must be properly cited. Students are strongly encouraged to refer to peer-reviewed research papers from reputed conferences and journals.

3. Minimum experimental requirements

The project should include:

1. implementation or reproduction of at least one meaningful PR method,
2. at least one baseline or comparative method,
3. at least one quantitative results table,
4. qualitative results wherever applicable,
5. analysis of at least one failure case,
6. clear interpretation of the obtained results.

Simply executing an existing GitHub repository without understanding, modifying, or analyzing the method will not be considered sufficient.

4. Submission Guidelines

Students should submit:

1. the final report in PDF format,
2. any additional supplementary results, if required.

Please note that files should be written and edited using L^AT_EX.

Compress the report and supplementary materials into zip folder.

Naming convention: <roll_number>_<name>.zip

5. Academic Integrity

All submitted work must reflect the student's own understanding. Code, figures, tables, equations, or text adapted from external sources must be appropriately acknowledged.

Use of open-source implementations is allowed for learning and experimentation, but students must clearly state which codebase was used, what components were implemented or modified, what experiments were performed independently, and what conclusions were drawn from those experiments.

Plagiarism, fabricated results, or submission of unmodified external reports/code as original work will be treated as academic misconduct.

The emphasis of the project is on understanding, mathematical clarity, implementation, experimentation, and critical analysis.